# Effective Minimum Wages & State Labor Force Participation Rates in the US (1990-2025)

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# Background

- Federal min. wage taken into effect since 1938 (Fair Labor Standards Act)
- Federal min. wage plateaued at $7.25/hr since 2009 despite increasing inflation
- Over 30 states now set their own min. wage above federal
- Complexity behind the policy:
  - Who is affected (age, skill level, gender, etc.)
  - Where it happens (state, sector of industry, etc.)
  - How much it increases over time

# Literature Review

- Most Vulnerable
  - **Young Adults & Low-Skilled Workers** → Larger employment losses; temporary *(Godoey and Reich 2021)*
  - Experience short-run disruption; effects fade within months
- Less Affected
  - **Prime-Age Workers (25-54)** → Minimal harm, often improved job quality *(Godoey and Reich 2021)*
  - **Higher-Skilled Workers** → Little to no negative effects
- Sectoral Patterns
  - **Food Service & Retail** → Wages rise, but hours may be cut *(Dube, Lester, and Reich 2010)*
  - **Manufacturing & High-Wage** → Minimal impact; fewer minimum-wage workers *(Yonezawa, et al. 2020)*

# Research Questions

What is the effect of minimum wage on the labor force participation rate (LFPR) at the state level?

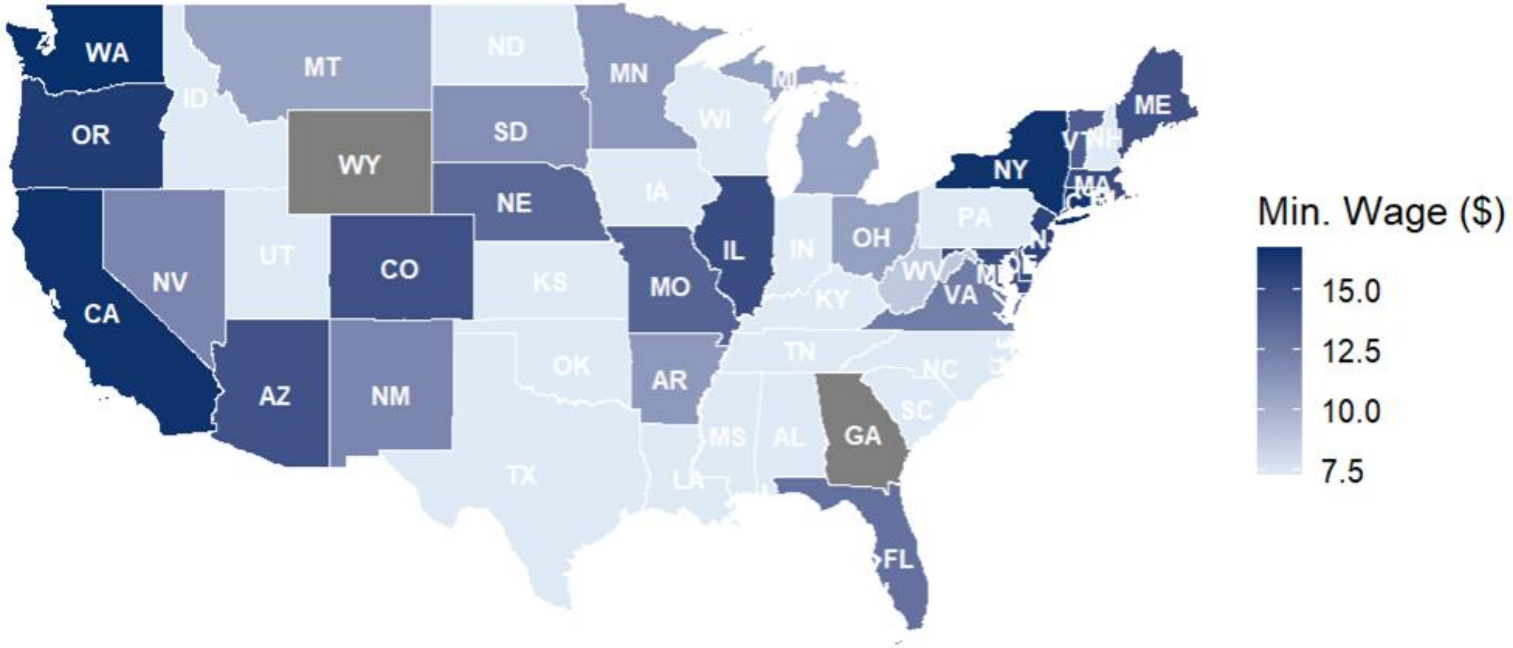
Does raising minimum wage reduce labor force participation, or do workers benefit?

# Data Collection

- API keys were acquired to collect:
  - **Federal and state minimum wage**, **unemployment**, and **LFPR**, using the **Fredr** package
  - **Median household income** and **education** data from census data using the **tidycensus** package
- Packages used in data processing: tidyverse, lubridate
- Variables:
  - Dependent variable is LFPR
  - Independent variable is effective minimum wage
  - Control variables include unemployment rate, median household income & educational attainment (at least bachelor's)
- Period is from beginning of 1990 to latest 2025 available data (Aug)

## Effective Minimum Wage by State (2025)

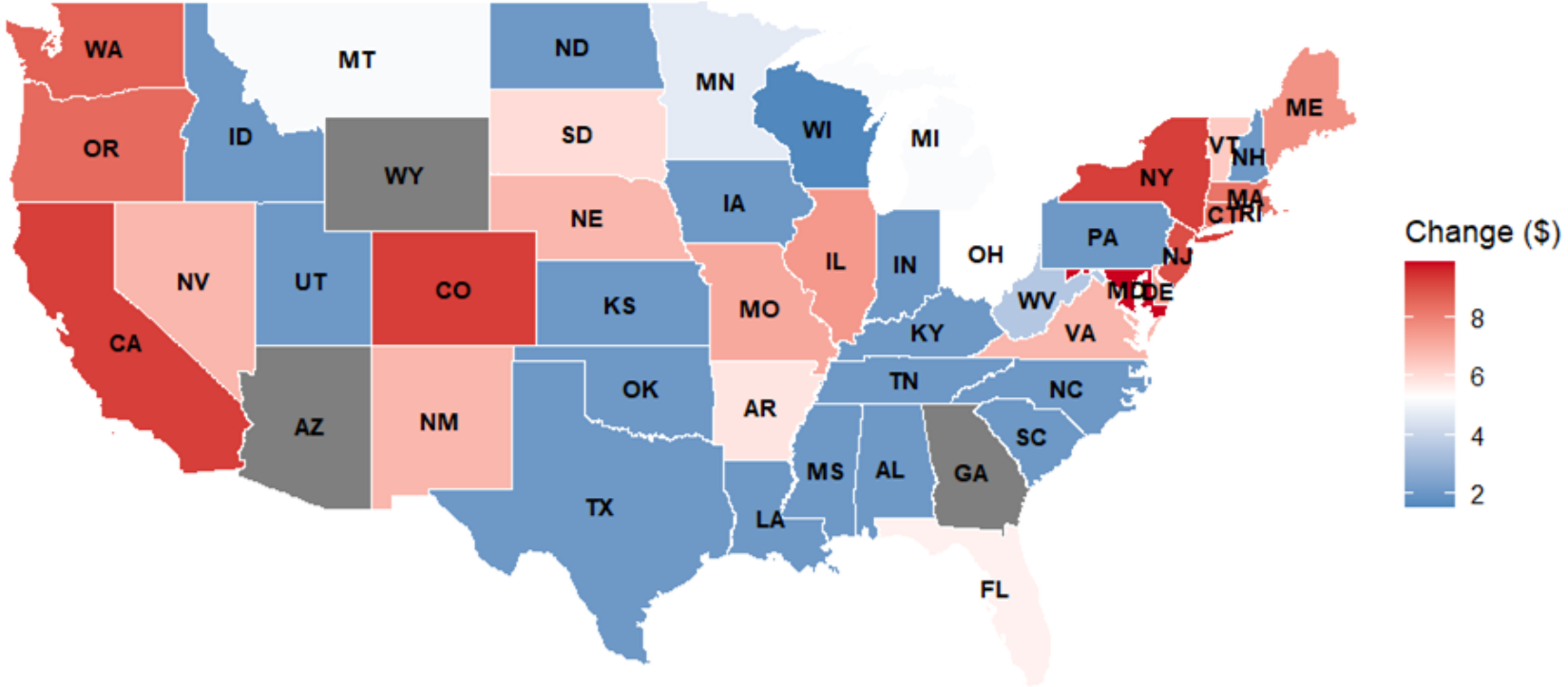
Higher minimum wage states appear in darker blue




Source: FRED, Census, State Policy Data

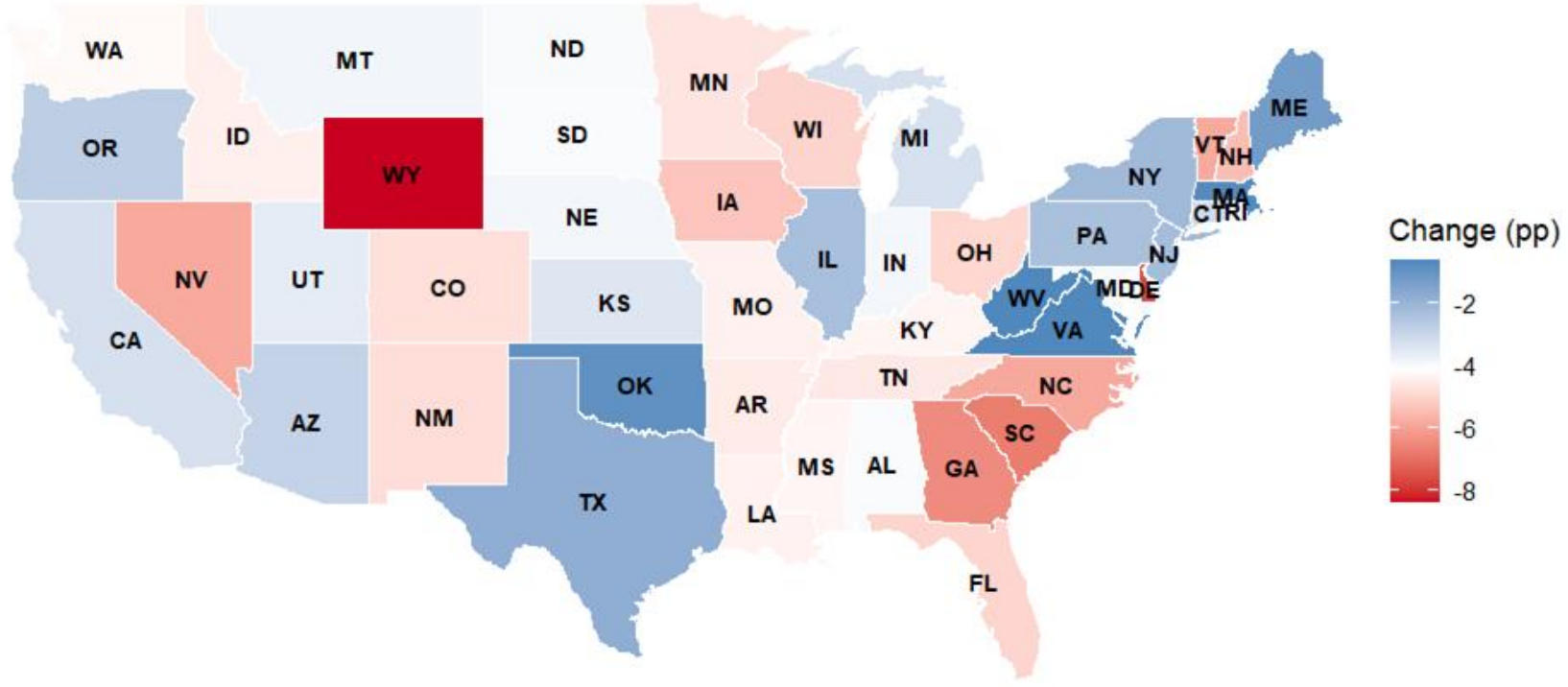
## Change in Minimum Wage (2006–2024)

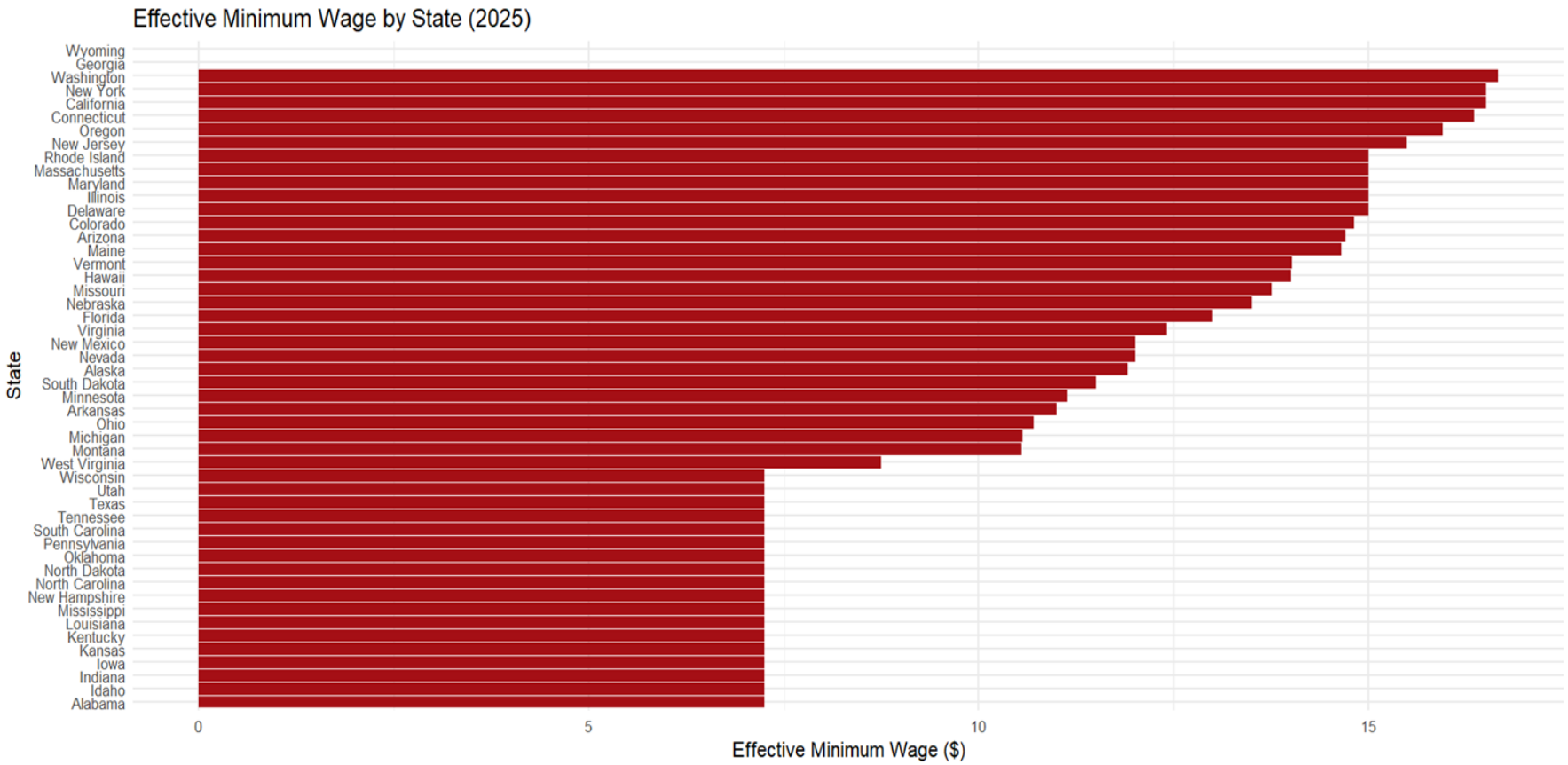
Blue = Small Increase, Red = Large Increase

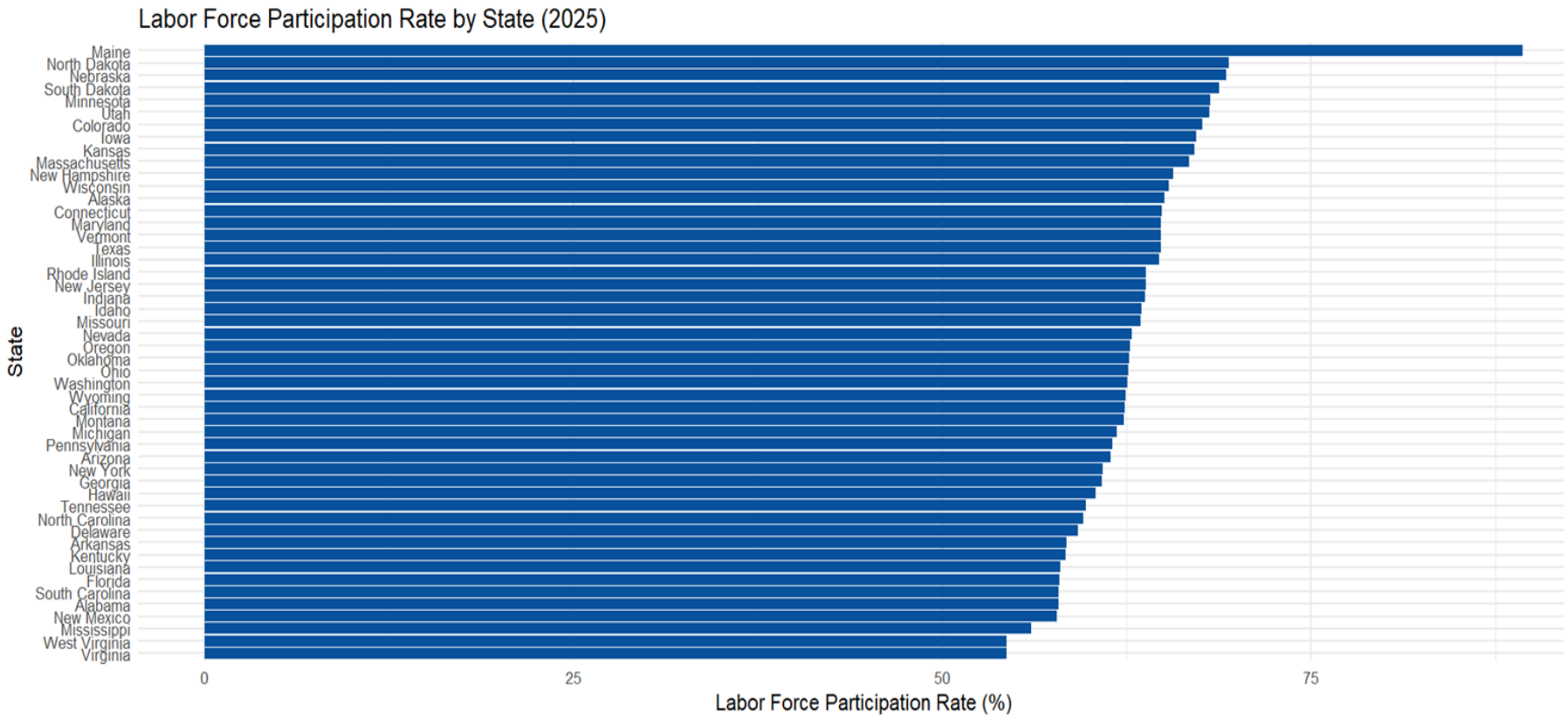


## Change in LFPR (2006–2024)

Red = Decline, Blue = Increase



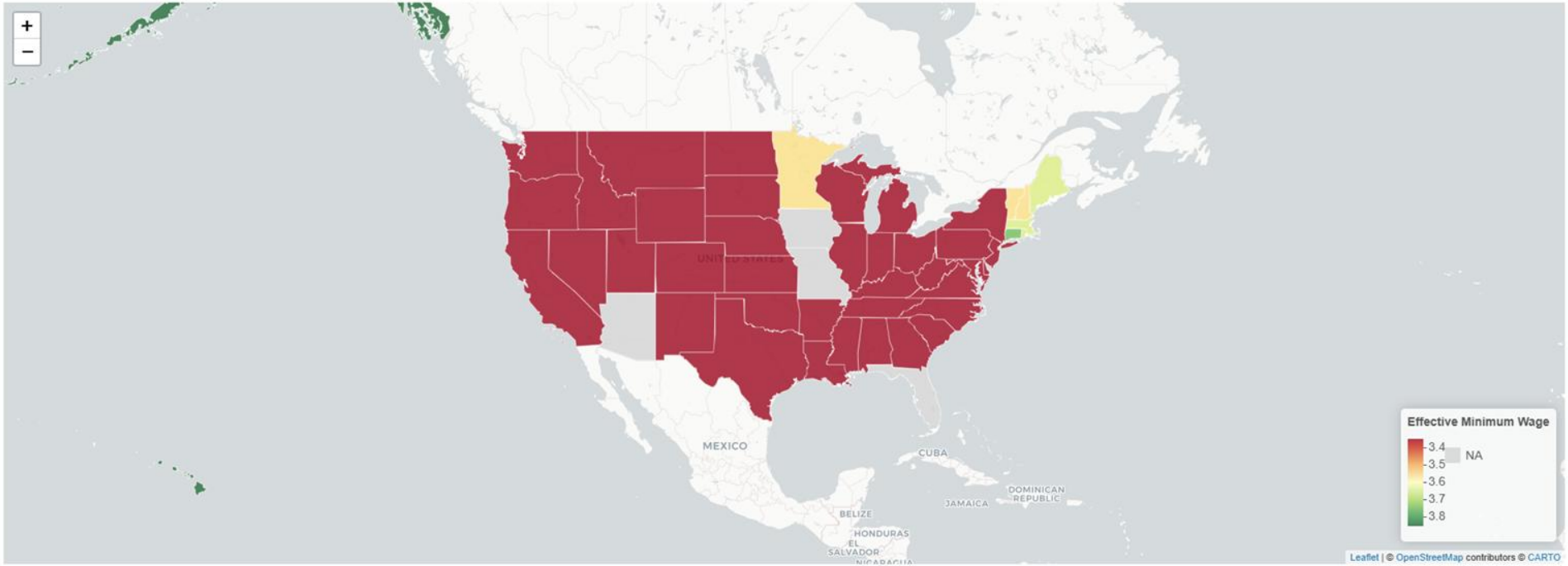

Effective Minimum Wage by State (2025)
State
Wyoming
Georgia
Washington
New York
California
Connecticut
Oregon
New Jersey
Rhode Island
Massachusetts
Maryland
Illinois
Delaware
Colorado
Arizona
Maine
Vermont
Hawaii
Missouri
Nebraska
Florida
Virginia
New Mexico
Nevada
Alaska
South Dakota
Minnesota
Arkansas
Ohio
Michigan
Montana
West Virginia
Wisconsin
Utah
Texas
Tennessee
South Carolina
Pennsylvania
Oklahoma
North Dakota
North Carolina
New Hampshire
Mississippi
Louisiana
Kentucky
Kansas
Iowa
Indiana
Idaho
Alabama
0
5
10
15
Effective Minimum Wage ($)


Labor Force Participation Rate by State (2025)
State
Maine
North Dakota
Nebraska
South Dakota
Minnesota
Utah
Colorado
Iowa
Kansas
Massachusetts
New Hampshire
Wisconsin
Alaska
Connecticut
Maryland
Vermont
Texas
Illinois
Rhode Island
New Jersey
Indiana
Idaho
Missouri
Nevada
Oregon
Oklahoma
Ohio
Washington
Wyoming
California
Montana
Michigan
Pennsylvania
Arizona
New York
Georgia
Hawaii
Tennessee
North Carolina
Delaware
Arkansas
Kentucky
Louisiana
Florida
South Carolina
Alabama
New Mexico
Mississippi
West Virginia
Virginia
0
25
50
75
Labor Force Participation Rate (%)

Standardized Trends of LFPR and Effective Minimum Wage (1990–2024)
Standardized Value (z-score)
2
1
0
-1
1990
2000
2010
2020
Year
Variable
Standardized LFPR
Standardized Effective Minimum Wage

Link to interactive map built using Shiny

# Map of Effective Minimum Wage & Labor Force Participation Rate in the US from 1990 to 2025

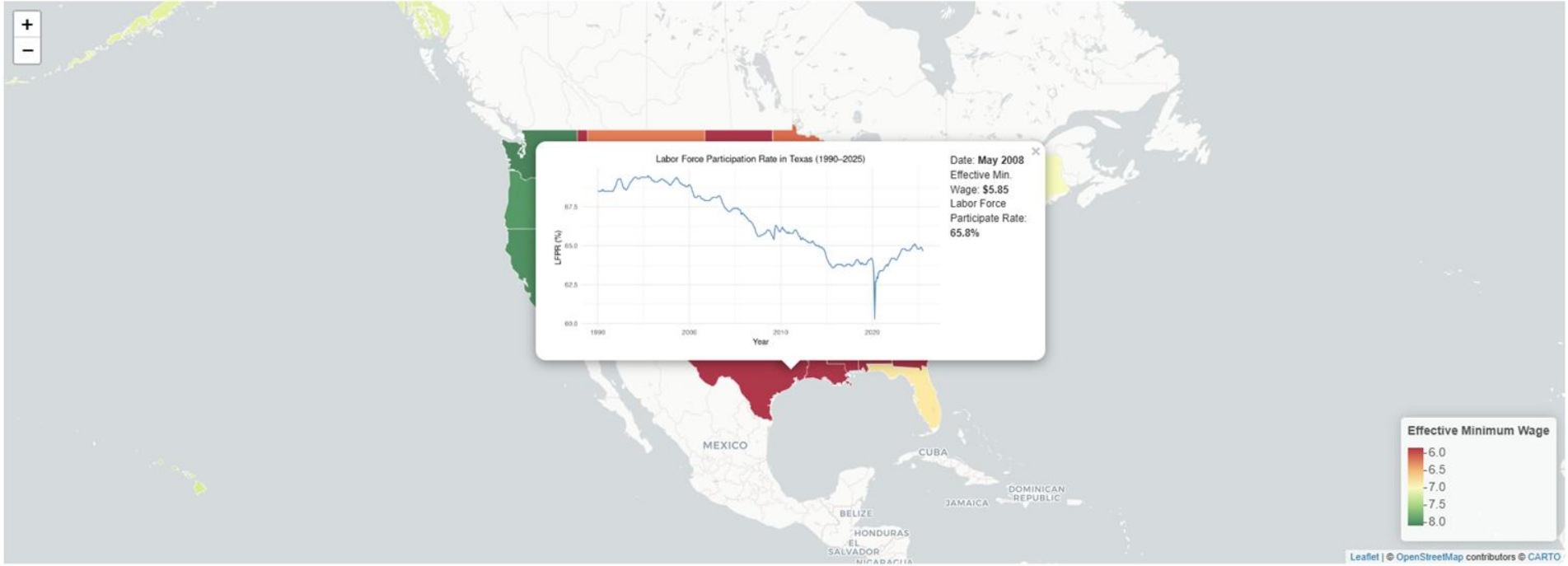



**Select month:**

1990-01-01 2025-08-01

1990-01-01 1997-02-13 2004-03-28 2011-05-11 2018-06-23 2025-08-01

# Map of Effective Minimum Wage & Labor Force Participation Rate in the US from 1990 to 2025

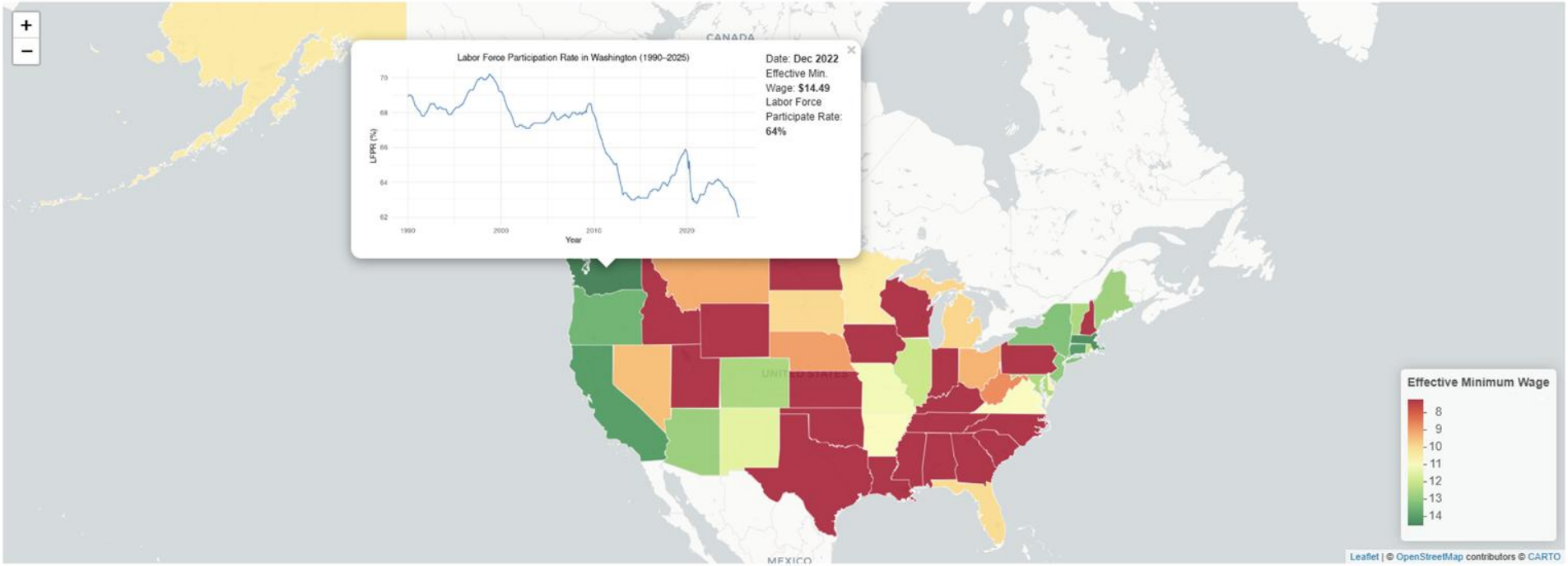



**Select month:**

1990-01-01 | 2008-04-21 | 2025-08-01

1990-01-01 1997-02-13 2004-03-28 2011-05-11 2018-06-23 2025-08-01

# Map of Effective Minimum Wage & Labor Force Participation Rate in the US from 1990 to 2025




**Select month:**

1990-01-01 2025-01-26

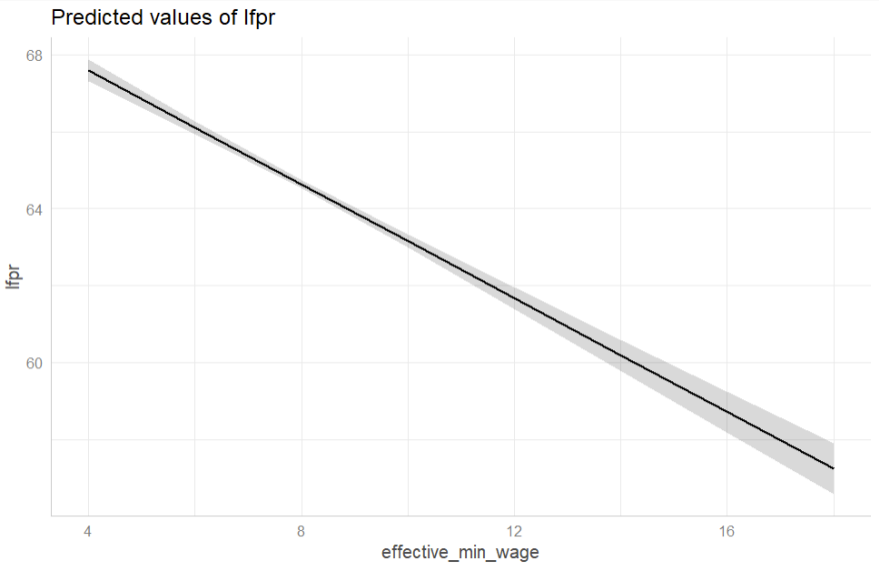
1990-01-01 1997-02-13 2004-03-28 2011-05-11 2018-06-23 2025-08-01

# Regression

OLS Method:

$$\text{LFPR}_{s,t} = \beta_0 + \beta_1 \text{MinWage}_{s,t} + \beta_2 \text{Unemp}_{s,t} + \beta_3 \text{Income}_{s,t} + \beta_4 \text{Edu}_{s,t} + \varepsilon_{s,t}$$

$s = \text{state}, \quad t = \text{time}$




- $1 increase in minimum wage roughly decreases the LFPR by 0.74 percentage point
- Could be biased (due to other external factors and doesn't take into account differences between states)
- $R^2$ is very low at just 0.071, meaning this model could only explain 7.1% of the variation

| Variable | Coeff. Est. | Std. Error | P-Value |
|---|---|---|---|
| Effective MinWage | -0.7398 | 0.03338 | <0.001 |
| Unemp | -0.1015 | 0.02528 | <0.001 |
| Median Income | -0.000004013 | 0.000001 | <0.001 |
| Education | 0.3090 | 0.01221 | <0.001 |

# Regression

One-Way Fixed Effects Method:

$$\mathrm{LFPR}_{s,t} = \beta_0 + \beta_1 \mathrm{MinWage}_{s,t} + \beta_2 \mathrm{Unemp}_{s,t} + \beta_3 \mathrm{Income}_{s,t} + \beta_4 \mathrm{Edu}_{s,t} + \text{State FE}_s + \varepsilon_{s,t}$$

$s = \text{state}$
$t = \text{time}$

$*\,\text{State FE} = \text{Value for a state at a given time} - \text{Avg value for that same state across all time}$

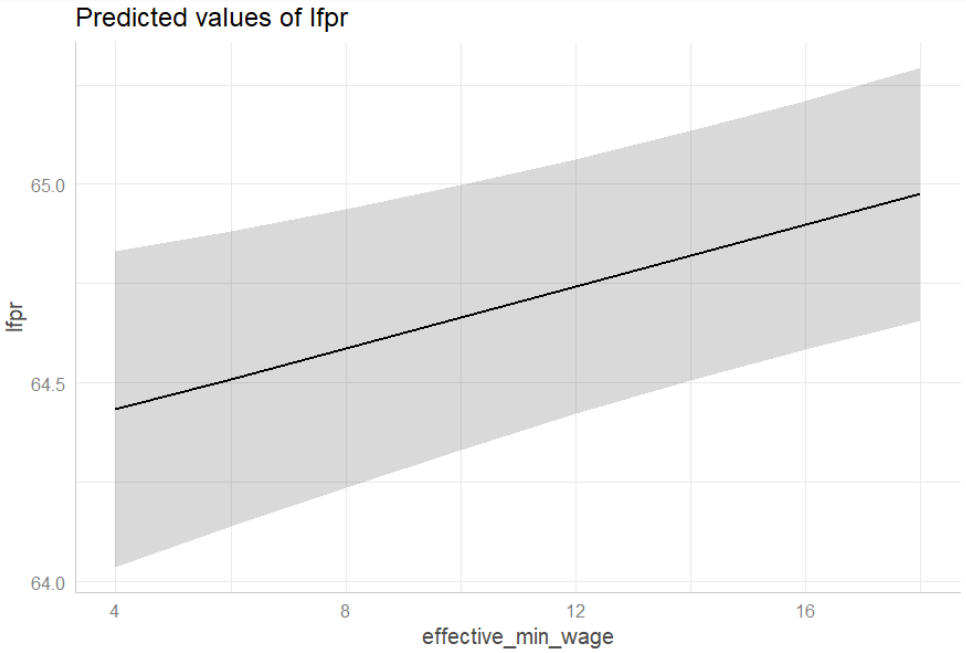
- Fixed effects help soften unobserved factors; i.e., demographics, economic shocks, etc. *(Wooldridge 2010)*
  - State FE takes each state and compare itself over time

| state | date | lfpr | effective_min_wage |
|---|---|---|---|
| Alabama | 1/1/1990 | 61.7 | 3.35 |
| Alabama | 2/1/1990 | 61.8 | 3.35 |
| Alabama | 3/1/1990 | 62 | 3.35 |
| Alabama | 4/1/1990 | 61.9 | 3.35 |
| Alabama | 5/1/1990 | 62.2 | 3.35 |
| Alabama | 6/1/1990 | 62.6 | 3.35 |
| Alabama | 7/1/1990 | 62.7 | 3.35 |
| Alabama | 8/1/1990 | 62.2 | 3.35 |
| Alabama | 9/1/1990 | 61.7 | 3.35 |
| Alabama | 10/1/1990 | 62.1 | 3.35 |
| Alabama | 11/1/1990 | 61.8 | 3.35 |
| Alabama | 12/1/1990 | 61.6 | 3.35 |
| Alabama | 1/1/1991 | 60.8 | 3.8 |

# Regression

- $1 increase in minimum wage roughly increase the LFPR by 0.039 percentage point within the same state
- $R^2$ is now higher at 0.5347, meaning the model can explain 53% of within-state variation
- Can still be biased because it leaves out economic shocks nationwide and other time trends
- Very wide CI because it looks at such a small variation

| Variable | Coeff. Est. | Std. Error | P-Value |
|---|---|---|---|
| Effective MinWage | 0.038701 | 0.011143 | 0.00052 |
| Unemp | -0.14051 | 0.007141 | <0.001 |
| Median Income | 0.00000175 | 0.0000002 | <0.001 |
| Education | -0.46244 | 0.006961 | <0.001 |



# Regression

R Codes for the Models:

```
model_ols <- lm(
  lfpr ~ effective_min_wage +
    unemployment_rate +
    median_income +
    education_ba_or_higher,
  data = df
)
```

```
model_fixed_effects_state <- plm(
  lfpr ~ effective_min_wage +
    unemployment_rate +
    median_income +
    education_ba_or_higher,
  data  = df_panel,
  model = "within",
  effect = "individual"
)
```

https://www.econometrics-with-r.org/10.3-fixed-effects-regression.html

# Limitations

- American Community Survey data (Census) for median household income and education only available from 2005 (also no data for 2020 due to COVID-19)
- Extremely low P-values possibly due to there being more than 10K+ observations in the models
- State FEs don’t account for all the unobserved variables
- Some data for some states across certain periods of time don’t get reported by the FRED

# References


- Godoey, Anna, and Michael Reich. 2021. “Are Minimum Wage Effects Greater in Low-Wage Areas?” *Industrial Relations* 60(1): 36–83. https://doi.org/10.1111/irel.12267.
- *Hanck, Christoph, Martin Arnold, Alexander Gerber, and Martin Schmelzer. 2019. “10.3 Fixed Effects Regression.”* Introduction to Econometrics with R. *Open Review edition. Accessed December 2, 2025. https://www.econometrics-with-r.org/10.3-fixed-effects-regression.html.*
- Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*. 2nd ed. Cambridge, MA: MIT Press, pp. 265–278.
- Yonezawa, Koichi, Miguel Ignacio Gomez, and Edward W. McLaughlin. 2020. “Impacts of Minimum Wage Increases in the US Retail Sector: Full-Time versus Part-Time Employment.” *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3520915.

# Thank you!